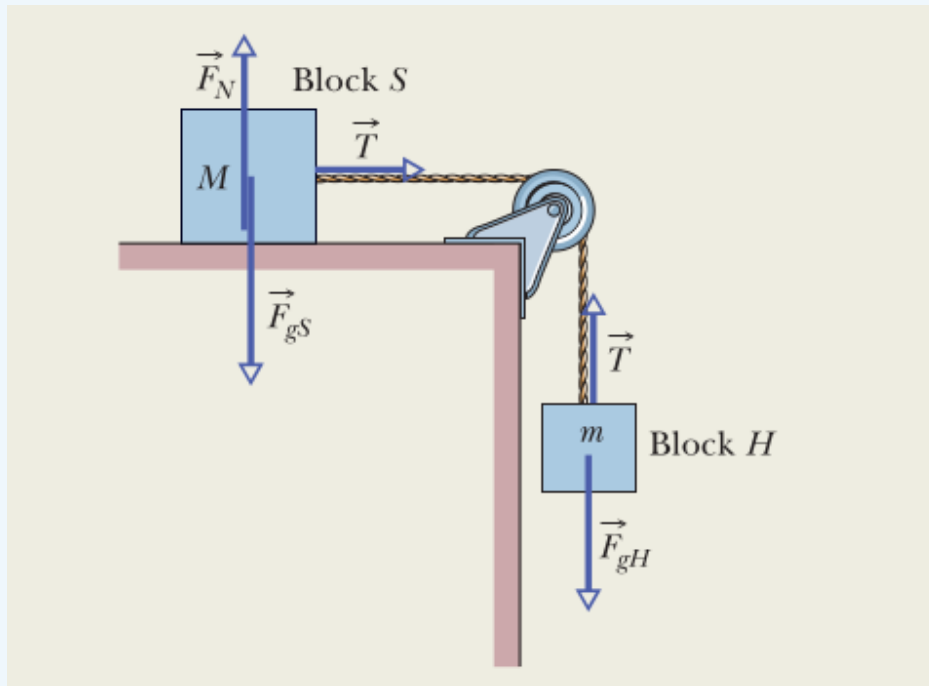


Mechanics - 2

- Problems and Solutions Based on Newton's
Laws

Problem 1

Figure shows a block S (the sliding block) with mass $M = 3.3$ kg. The block is free to move along a horizontal frictionless surface and connected, by a cord that wraps over a frictionless pulley, to a second block H (the hanging block), with mass $m = 2.1$ kg. The cord and pulley have negligible masses compared to the blocks (they are "massless"). The hanging block H falls as the sliding block S accelerates to the right.



Find:

- (a) The acceleration of block S .
- (b) The acceleration of block H .
- (c) The tension in the cord.

Solution

Let the acceleration of both blocks be a . Since the blocks are connected by a cord, they have the same magnitude of acceleration.

Equations of Motion

For block S (sliding block): Along x-axis,

$$T = Ma$$

Along y-axis,

$$F_N - F_g = 0$$

For block H (hanging block):

$$mg - T = ma$$

Acceleration

Substituting $T = Ma$ into the equation for block H :

$$mg - Ma = ma$$

$$mg = Ma + ma$$

$$a = \frac{mg}{M + m}$$

Substituting the given values:

$$a = \frac{(2.1)(9.8)}{3.3 + 2.1} = \frac{20.58}{5.4} = 3.81 \text{ m/s}^2$$

Tension

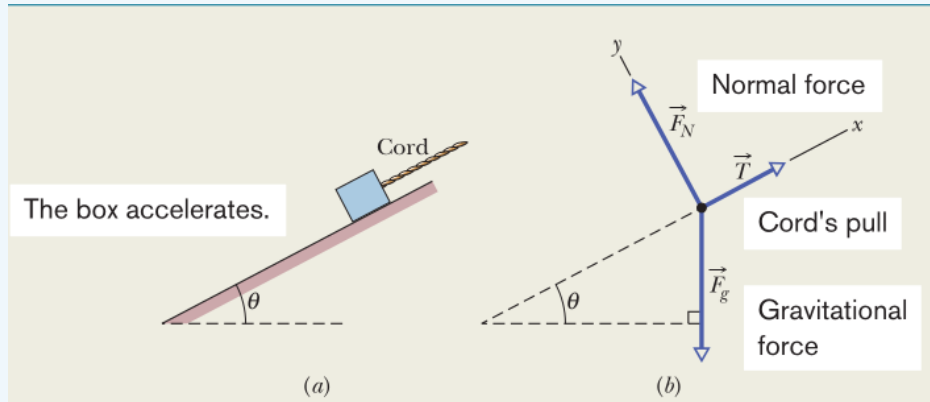
$$T = Ma = (3.3)(3.81) = 12.57 \text{ N}$$

Answers

- (a) Acceleration of block S is 3.81 m/s^2 .
- (b) Acceleration of block H is 3.81 m/s^2 .
- (c) Tension in the cord is 12.57 N .

Problem 2

In Figure, a cord pulls a box of biscuits up along a frictionless plane inclined at angle $\theta = 30.03^\circ$. The box has mass $m = 5.00$ kg, and the force from the cord has magnitude $T = 25.0$ N.



- (a) What is the box's acceleration a along the inclined plane?
- (b) Find normal force F_N .

Solution

Equations of Motion

Applying Newton's Second Law along the incline:

Along X axis:

$$T - mg \sin \theta = ma$$

Along Y axis:

$$F_N - mg \cos \theta = 0$$

Acceleration

$$a = \frac{T - mg \sin \theta}{m}$$

Substituting the given values:

$$a = \frac{25.0 - (5.00)(9.8) \sin 30.03^\circ}{5.00} = 0.098 \text{ m/s}^2$$

Normal Force

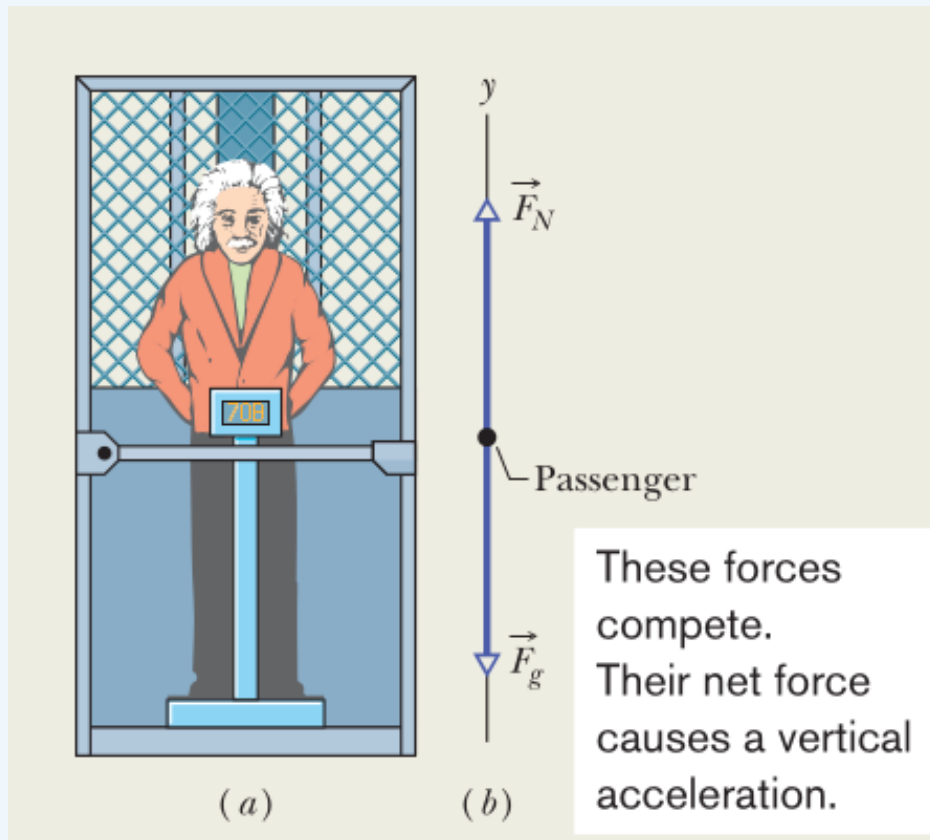
$$F_N = mg \cos \theta = (5.00)(9.8) \cos 30.03^\circ = 42.47 \text{ N}$$

Answers

- Box acceleration along the inclined plane: 0.098 m/s^2 .
- Normal force: 42.47 N .

Problem 3

A passenger of mass $m = 72.2$ kg stands on a platform scale in an elevator cab. We are concerned with the scale readings when the cab is stationary and when it is moving up or down.



- (a) Find a general solution for the scale reading, whatever the vertical motion of the cab.
- (b) What does the scale read if the cab is stationary or moving upward at a constant 0.50 m/s?
- (c) What does the scale read if the cab accelerates upward at 3.20 m/s² and downward at 3.20 m/s²?

Solution

(a) General Solution

The platform scale measures the normal force N exerted on the passenger. Applying Newton's Second Law:

For upward motion:

$$F_N - mg = ma \implies F_N = m(g + a)$$

For downward motion:

$$F_N - mg = -ma \implies F_N = m(g - a)$$

where $g = 9.8 \text{ m/s}^2$, and a is the acceleration of the elevator.

(b) Scale Reading when the Cab is Stationary or Moving at Constant Velocity

$$F_N = mg = (72.2)(9.8) = 707.6 \text{ N}$$

(c) Scale Reading when the Cab Accelerates

Upward acceleration:

$$F_N = 72.2(9.8 + 3.2) = 938.6 \text{ N}$$

Downward acceleration:

$$F_N = 72.2(9.8 - 3.2) = 476.5 \text{ N}$$

Answers

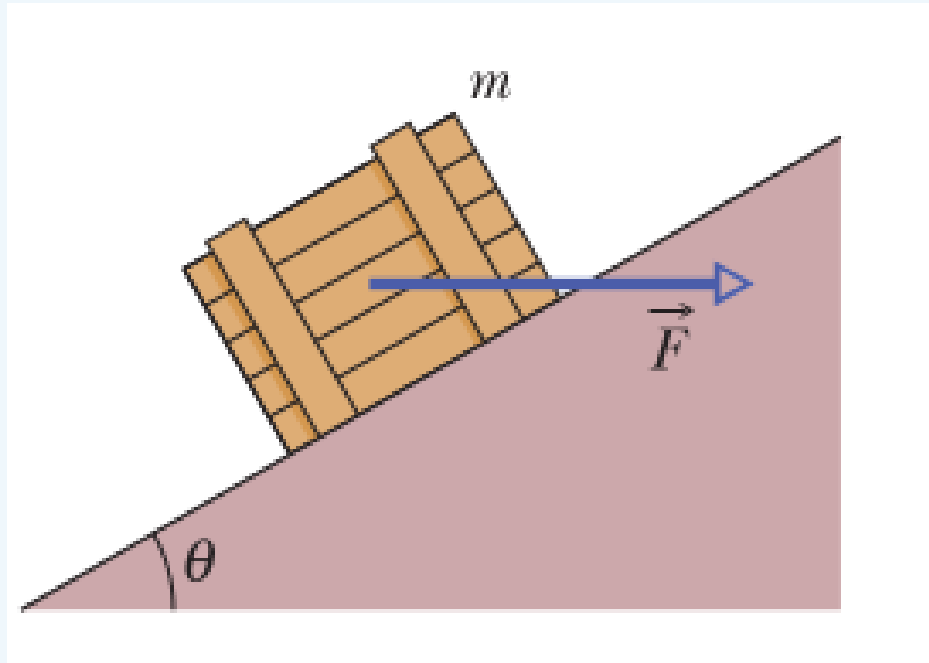
(a) General solution: $F_N = m(g \pm a)$.

(b) Scale reading when stationary or moving at constant speed: 707.6 N.

(c) Scale reading when accelerating upward: 938.6 N, downward: 476.5 N.

Problem 4

A crate of mass $m = 100 \text{ kg}$ is pushed at constant speed up a frictionless ramp with an incline angle of $\theta = 30.0^\circ$ by a horizontal force F .



- (a) What is the magnitude of F ?
- (b) What is the magnitude of the force on the crate from the ramp?

Solution

(a) Horizontal Force

For constant speed, acceleration $a = 0$. Along the incline:

$$F \cos \theta = mg \sin \theta \implies F = \frac{mg \sin \theta}{\cos \theta} = mg \tan \theta$$

Substituting values:

$$F = 100 \times 9.8 \times \tan 30^\circ \approx 100 \times 9.8 \times 0.577 = 565.5 \text{ N}$$

(b) Force from the Ramp (Normal Force)

$$F_N = mg \cos \theta + F \sin \theta = 100 \times 9.8 \times \cos 30^\circ + 565.5 \times \sin 30^\circ = 1131 \text{ N}$$

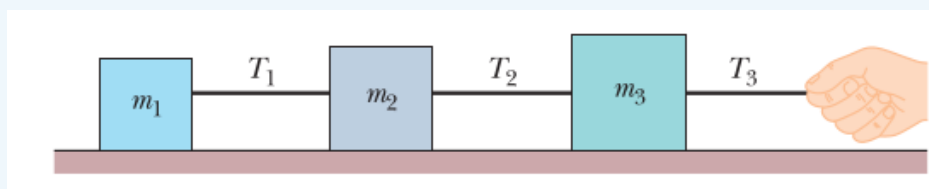
Answers

- (a) Horizontal force: $F \approx 565.5 \text{ N}$
- (b) Force from ramp: $F_N \approx 848.7 \text{ N}$

Problem 5

Three connected blocks are pulled to the right on a horizontal frictionless table by a force of magnitude $T_3 = 65.0 \text{ N}$. The masses of the blocks are:

- $m_1 = 12.0 \text{ kg}$,
- $m_2 = 24.0 \text{ kg}$,
- $m_3 = 31.0 \text{ kg}$.



Calculate:

- The magnitude of the system's acceleration,
- The tension T_1 (force exerted by m_2 on m_1),
- The tension T_2 (force exerted by m_3 on m_2).

Solution

(a) Calculation of Acceleration

Total mass:

$$m_{\text{total}} = m_1 + m_2 + m_3 = 12.0 + 24.0 + 31.0 = 67.0 \text{ kg}$$

Force equation:

$$F = m_{\text{total}}a \implies a = \frac{F}{m_{\text{total}}} = \frac{65.0}{67.0} \approx 0.970 \text{ m/s}^2$$

(b) Tension T_2

$$T_2 = (m_1 + m_2)a = (12.0 + 24.0) \times 0.970 = 34.9 \text{ N}$$

(c) Tension T_1

$$T_1 = m_1a = 12.0 \times 0.970 = 11.6 \text{ N}$$

Answers

- (a) System acceleration: 0.970 m/s^2
- (b) Tension $T_2 = 34.9 \text{ N}$
- (c) Tension $T_1 = 11.6 \text{ N}$

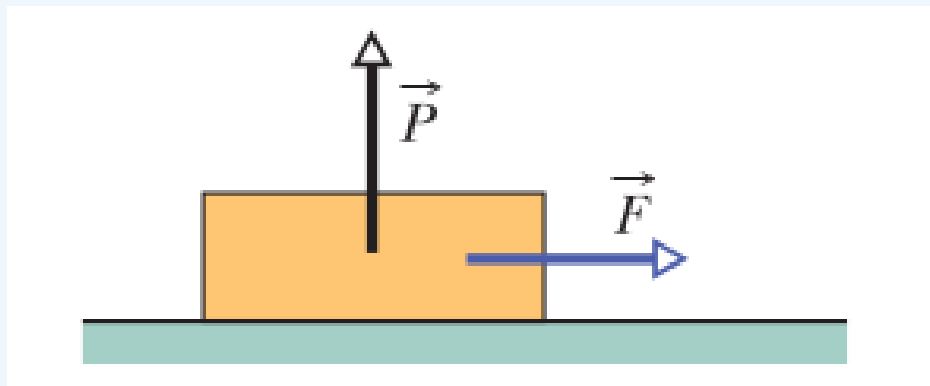
Problem 6

A 2.5 kg block is initially at rest on a horizontal surface. A horizontal force F of magnitude 6.0 N and a vertical force P are then applied to the block. The coefficients of friction for the block and surface are:

- Static friction coefficient: $\mu_s = 0.40$,
- Kinetic friction coefficient: $\mu_k = 0.25$.

Determine the magnitude of the frictional force acting on the block if the magnitude of P is:

- (a) 8.0 N,
- (b) 10.0 N,
- (c) 12.0 N.



Solution

The normal force is reduced by the vertical force P :

$$F_N = mg - P$$

Maximum static friction:

$$f_s^{\max} = \mu_s F_N$$

Case (a) $P = 8.0 \text{ N}$:

$$F_N = 2.5 \cdot 9.8 - 8.0 = 16.5 \text{ N}, \quad f_s^{\max} = 0.4 \cdot 16.5 = 6.6 \text{ N}$$

Case (b) $P = 10.0 \text{ N}$:

$$F_N = 2.5 \cdot 9.8 - 10.0 = 14.5 \text{ N}, \quad f_s^{\max} = 0.4 \cdot 14.5 = 5.8 \text{ N}$$

As the maximum static frictional force is less than the applied force, block will start moving. So the frictional force will be kinetic.

$$f_k = \mu_k F_N = 0.25 \times 14.5 = 3.625 \text{ N}$$

Case (c) $P = 12.0 \text{ N}$:

$$F_N = 2.5 \cdot 9.8 - 12.0 = 12.5 \text{ N}, \quad f_s^{\max} = 0.4 \cdot 12.5 = 5.0 \text{ N}$$

Since $F = 6.0 \text{ N}$, the block moves when $F > f_s^{\max}$, and kinetic friction applies:

$$f_k = \mu_k F_N = 0.25 \times 12.5 = 3.125 \text{ N}$$

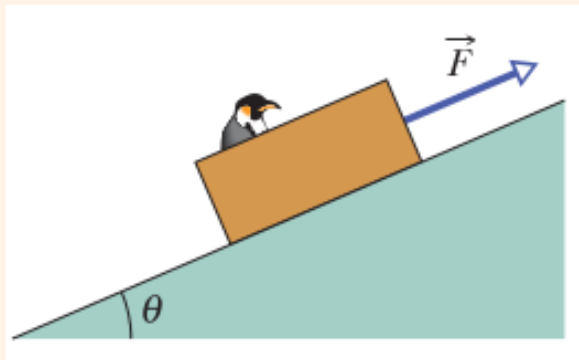
Answers

- (a) Frictional force (Static): 6.6 N
- (b) Frictional force (Kinetic): 3.625 N (block starts moving)
- (c) Frictional force (Kinetic): 3.125 N (block starts moving)

Problem 7

A loaded penguin sled weighing 80 N rests on a plane inclined at an angle $\theta = 20^\circ$ to the horizontal. Between the sled and the plane, the coefficient of static friction is $\mu_s = 0.25$, and the coefficient of kinetic friction is $\mu_k = 0.15$.

- (a) What is the least magnitude of the force F , parallel to the plane, that will prevent the sled from slipping down the plane?
- (b) What is the minimum magnitude F that will start the sled moving up the plane?
- (c) What value of F is required to move the sled up the plane at constant velocity?



Solution 7

Let the weight of the sled be $W = 80 \text{ N}$, and the angle of inclination be $\theta = 20^\circ$. The normal force F_N acting on the sled is given by:

$$F_N = W \cos \theta = 80 \text{ N} \cdot \cos(20^\circ) \approx 75.08 \text{ N}$$

(a) Preventing the sled from slipping down the plane

The maximum static frictional force is:

$$f_s = \mu_s F_N = 0.25 \cdot 75.08 \approx 18.77 \text{ N}$$

The component of the weight acting down the plane is:

$$W_{\parallel} = W \sin \theta = 80 \cdot \sin(20^\circ) \approx 27.47 \text{ N}$$

Minimum force to prevent slipping:

$$F \geq W_{\parallel} - f_s = 27.47 - 18.77 \approx 8.70 \text{ N}$$

(b) Minimum force to start moving up the plane

$$F_{\min} = W_{\parallel} + f_s = 27.47 + 18.77 \approx 46.24 \text{ N}$$

(c) Force for constant velocity

The kinetic frictional force:

$$f_k = \mu_k F_N = 0.15 \cdot 75.08 \approx 11.26 \text{ N}$$

Total force:

$$F_{\text{constant velocity}} = W_{\parallel} + f_k = 27.47 + 11.26 \approx 38.73 \text{ N}$$

Answers 7

(a) $F \approx 8.70 \text{ N}$

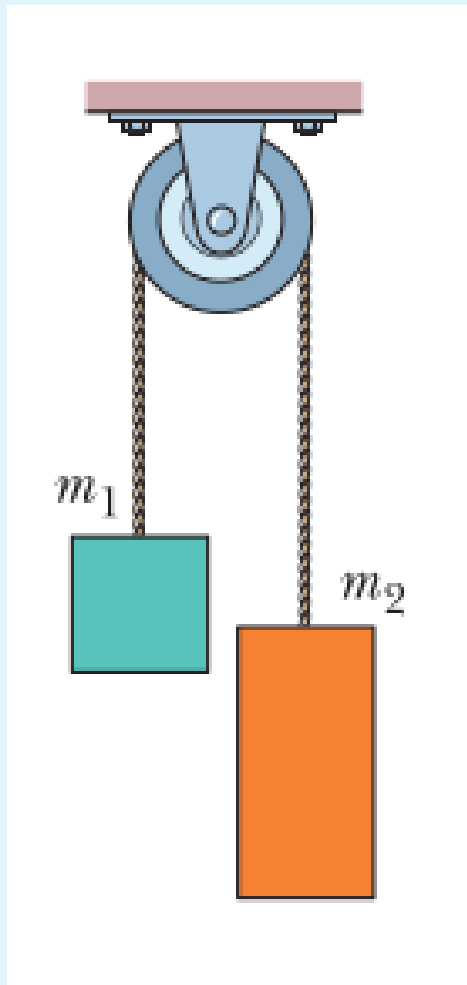
(b) $F_{\min} \approx 46.24 \text{ N}$

(c) $F_{\text{constant velocity}} \approx 38.73 \text{ N}$

Problem 8

Two blocks are connected by a cord (of negligible mass) that passes over a frictionless pulley (also of negligible mass). This arrangement is known as Atwood's machine. One block has mass $m_1 = 1.30$ kg; the other has mass $m_2 = 2.80$ kg.

- (a) What is the magnitude of the blocks' acceleration?
- (b) What is the tension in the cord?



Solution 8

Let the acceleration of the blocks be a and the tension in the cord be T . Applying Newton's second law for each block:

Block 1 (mass m_1)

$$T - m_1g = m_1a$$

Block 2 (mass m_2)

$$m_2g - T = m_2a$$

Adding these equations to eliminate T :

$$m_2g - m_1g = (m_1 + m_2)a$$

$$a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(2.80 - 1.30) \cdot 9.8}{1.30 + 2.80} \approx 3.59 \text{ m/s}^2$$

Tension in the cord

$$T = m_1g + m_1a = 1.30 \cdot 9.8 + 1.30 \cdot 3.59 \approx 17.44 \text{ N}$$

Answers 8

(a) Acceleration: $a \approx 3.59 \text{ m/s}^2$

(b) Tension: $T \approx 17.44 \text{ N}$